

Computing Matrix Rank Using Echelon Form

High-Level Overview

We have learned how to reduce a matrix into **Row Echelon Form (REF)** using elementary operations. Now, we connect this technique directly to the **rank of a matrix**.

The Core Principle

Instead of calculating complex determinants of minors, engineering mathematics allows us to define the **row rank** as the exact number of non-zero rows present in the matrix's row echelon form.

- **Theorem:** The row rank of a matrix is always equal to its column rank, and this common value is called simply the **rank of the matrix** ($\rho(A)$).
- **Linear Independence:** Conceptually, the rank tells us the **maximum number of linearly independent row or column vectors** inside that matrix.
- **Important Bounds:** For any $m \times n$ matrix A , the rank is bounded by the smaller dimension: $\rho(A) \leq \min(m, n)$.

Step-by-Step Worked Example 1: Finding Rank via REF

Let's compute the rank of matrix A by transforming it into row echelon form.

Given Matrix:

$$A = \begin{bmatrix} 2 & -1 & 3 \\ 1 & 0 & 1 \\ 0 & 2 & -1 \\ 1 & 1 & 4 \end{bmatrix}$$

Step 1: Swap Rows to Get a Clean Pivot

Swap row 1 and row 2 to place a 1 at the top-left position:

- Operation: $R_1 \leftrightarrow R_2$

$$\sim \begin{bmatrix} 1 & 0 & 1 \\ 2 & -1 & 3 \\ 0 & 2 & -1 \\ 1 & 1 & 4 \end{bmatrix}$$

Step 2: Clear Out Column 1 Under the Pivot

Perform row operations to eliminate entries below the leading 1:

- Operations: $R_2 \rightarrow R_2 - 2R_1$ and $R_4 \rightarrow R_4 - R_1$

$$\sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & -1 & 1 \\ 0 & 2 & -1 \\ 0 & 1 & 3 \end{bmatrix}$$

Step 3: Normalize the Second Row Pivot

Scale the second row to make the leading entry positive:

- Operation: $R_2 \rightarrow (-1)R_2$

$$\sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 2 & -1 \\ 0 & 1 & 3 \end{bmatrix}$$

Step 4: Continue Down the Staircase

Clear entries below the second row pivot:

- Operations: $R_3 \rightarrow R_3 - 2R_2$ and $R_4 \rightarrow R_4 - R_2$

$$\sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \\ 0 & 0 & 4 \end{bmatrix}$$

Step 5: Finalize Row Echelon Form

Eliminate the last remaining entry under the third pivot:

- Operation: $R_4 \rightarrow R_4 - 4R_3$

$$\sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Determining the Rank:

Count the number of non-zero rows remaining in this row echelon form. There are **3** non-zero rows.

- Thus, the rank of matrix A is **3**, written as $\rho(A) = 3$.

Step-by-Step Worked Example 2: Matrix with Linearly Dependent Rows

Let's find the rank of a 4×4 matrix B :

$$B = \begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}$$

Reduction Steps:

1. **Swap rows** to put a non-zero leading entry at the top: $R_1 \leftrightarrow R_2$.
2. **Eliminate entries** in column 1: $R_3 \rightarrow R_3 - 3R_1$ and $R_4 \rightarrow R_4 - R_1$.
3. **Clear subsequent rows** using the second pivot: $R_3 \rightarrow R_3 - R_2$ and $R_4 \rightarrow R_4 - R_2$.

The resulting REF matrix becomes:

$$\sim \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -3 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Determining the Rank:

Notice that two entire rows have collapsed into zeros. Only **2** non-zero rows remain.

- Therefore, $\rho(B) = 2$.

Summary Checklist for Rapid Revision

- ☐ **Definition:** The rank of a matrix equals the total count of non-zero rows in its **Row Echelon Form**.
- ☐ **Notation:** Represented as $\rho(A)$.
- ☐ **Constraint:** For an $m \times n$ matrix, $\rho(A) \leq \min(m, n)$, meaning rank can never exceed the smaller dimension.
- ☐ **System Application:** A linear system $AX = B$ is consistent (has a solution) if and only if the rank of coefficient matrix A equals the rank of the augmented matrix $[A|B]$.